

KALYAN MYSKER

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SUMMARY

Computer Science student focused on distributed systems, high-performance networking, multi-threading and secure communication protocols. Combines hands-on experience in cloud infrastructure with performance engineering and algorithmic optimization. Seeking Software Engineering internships to solve scalable infrastructure challenges.

EDUCATION

IIIT Basar (RGUKT), Telangana

Expected May 2028

Bachelor of Technology in Computer Science and Engineering

GPA: 8.0/10.0

Coursework: Data Structures, Algorithms, DBMS, Operating Systems, Computer Networks

TECHNICAL SKILLS

- **Languages:** Python, Java, SQL, C.
- **Cloud Platforms:** AWS (EC2, S3, Lambda).
- **DevOps & Tools:** CI/CD, GitHub Actions, Ansible, Docker, Git, Linux/Unix, Kubernetes (Basics).
- **Frameworks & Databases:** FastAPI, Pydantic, MySQL.
- **Testing:** pytest.
- **Core Concepts:** Distributed Systems, Object-Oriented Programming (OOP), RESTful APIs, Multi-threading, Load Balancing, Networking, High Availability, DSA.

TECHNICAL PROJECTS

UPI Offline Mesh Simulator

Python, FastAPI, Pydantic, RSA/AES, Docker

- Architected a decentralized payment gateway using custom mesh-routing, increasing transaction reach by 100% by implementing a high-integrity hybrid encryption pipeline (RSA-OAEP + AES-GCM).
- Ensured system reliability by implementing an atomic idempotency layer in a FastAPI backend using Pydantic models with automated testing using pytest.

Deep Packet Inspection (DPI) Engine

Python, PCAP, Networking

- Architected a high-performance DPI engine capable of real-time traffic classification and filtering, reducing unapproved network traffic by 90% via custom flow-tracking algorithms.
- Engineered a scalable packet processing pipeline to support offline PCAP analysis and live network sniffing with multithreaded architecture.

VectorDB Search Engine

Python, FastAPI, HNSW, RAG

- Engineered a custom vector database with a RAG pipeline, improving semantic retrieval efficiency for 768D embeddings by implementing HNSW and KD-Tree algorithms.
- Designed a high-throughput REST API using FastAPI to enable real-time document chunking and semantic search.

EXPERIENCE

Independent Software Development Projects

- Engineered complex distributed systems with a core focus on high-performance networking, cryptographic security protocols, and system-level performance optimization using Python.
- Implemented modular software architectures and containerized CI/CD pipelines, utilizing Ansible for configuration management, Docker for isolated environments, and GitHub Actions for continuous deployment automation.

ACHIEVEMENTS

- **Algorithmic Proficiency:** Solved 300+ LeetCode problems, focusing on advanced data structures, dynamic programming, and algorithmic optimization.
- **Certifications:** Oracle Cloud Infrastructure DevOps Professional, AWS Cloud Essentials, and Networking Core Knowledge Badges.
- **Job Simulations:** AWS / Forage Solutions Architecture Virtual Experience.